Software Requirements Specification

Campus Grievance Addressal System

Version 1.0

Prepared by

Group Number: II-8

Soham Siddhartha Roy	**25CSB0A22**
Jibendra Kumar Garai	**25CSB0A12**
Chaitanya Sathone	**25CSB0A54**
Aniket Rajesh Borkar	**25CSB0A75**

Project Owner:	*Not assigned*
Course:	**CS1207 Object Oriented Programming Lab**
Date:	**11/10/2026**

This template is based on the one available from the GMU site by Dr. Rob Pettit. Modifications will be used for academic purposes only.

Contents

1 Introduction 3
1.1 Document Purpose 3
1.2 Product Scope 3
1.3 Definitions, Acronyms and Abbreviations 3
1.4 Document Conventions 4
1.5 References and Acknowledgments 4
2 Overall Description 5
2.1 Product Overview 5
2.2 Product Functionality 5
2.3 Design and Implementation Constraints 5
2.4 Assumptions and Dependencies 6
3 Specific Requirements 7
3.1 Functional Requirements 7
3.2 Use Case Model 8
3.2.1 Use Case #1 (Login – U1) 8
3.2.2 Use Case #2 (Change Password – U2) 9
3.2.3 Use Case #3 (Raise Grievance – U3) 9
3.2.4 Use Case #4 (Track Tickets – U4) 10
3.2.5 Use Case #5 (View and Sort Ticket Queue – U5) 11
3.2.6 Use Case #6 (Update Ticket – U6) 11
3.2.7 Use Case #7 (Give Feedback or Reopen – U7) 12
3.2.8 Use Case #8 (Manage Records and Categories – U8) 12
3.2.9 Use Case #9 (View Logs and Reports – U9) 13
4 Other Non-functional Requirements 14
4.1 Performance Requirements 14
4.2 Safety and Security Requirements 14
5 Other Requirements 14
Appendix A - Activity Log 14

Revisions

Version	Primary Author(s)	Description of Version	Date Completed
Draft 1	Soham, Jibendra, Chaitanya, Aniket	First complete draft of the SRS	11/10/2026

1 Introduction

The Campus Grievance Addressal System (CGAS) replaces scattered complaint channels with one portal where students file grievances and administration tracks and resolves them. This section states the purpose and scope of the document and defines the terms used.

1.1 Document Purpose

This SRS specifies the software requirements for CGAS, version 1.0. It covers the complete system: the student portal, the administration portal, and the shared ticket back end.

It is written for the development team, the project owner (TA) and the course instructor. It is the reference for design, implementation and testing.

1.2 Product Scope

CGAS is a web-based portal that accepts every kind of campus grievance (plumbing, carpentry, electrical, masonry, administrative and security) through one form. Each complaint becomes a categorised ticket with a status and timestamp that both students and administrators can see.

Benefits:

- Students use one place for every complaint instead of visiting different offices.
- Administrators get a single queue, sorted by time, instead of separate registers.
- Duplicate complaints about the same fault are detected and merged, so staff do not repeat work.
- Every ticket has a recorded history, which makes resolution time measurable.

Objectives: reduce average resolution time, remove duplicate work, and give students visibility into progress.

1.3 Definitions, Acronyms and Abbreviations

Term	Meaning
Admin	A user of the administration portal who views and resolves tickets
Category	The grievance type: Plumbing, Carpentry, Electrical, Masonry, Administrative or Security
CGAS	Campus Grievance Addressal System
Deduplication	Detecting that a new ticket describes the same issue as an open ticket and linking them
Grievance	A complaint or request about a campus facility or service
OTP	One-Time Password
SRS	Software Requirements Specification
Staff member	A maintenance or administrative worker to whom an admin assigns tickets; staff are held as records and do not log in
Student	A registered campus resident or enrolled student who raises tickets
Ticket	The record created for one grievance, with category, status, location and timestamps

UI	User Interface

1.4 Document Conventions

- Arial font, size 11 for body text, single spacing and 1 inch margins, following IEEE formatting.
- Italics are used for comments and placeholders.
- Functional requirements are numbered F1, F2 and so on. Use cases are numbered U1, U2 and so on.
- "Shall" marks a mandatory requirement.
- Ticket statuses are written in capitals when referred to: OPEN, IN PROGRESS, RESOLVED, CLOSED, DUPLICATE.

1.5 References and Acknowledgments

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
- GMU SRS template by Dr. Rob Pettit, as supplied for CS1207 Object Oriented Programming Lab.
- Sample SRS: Mess Fee Management System, supplied by the course for reference.
- UML 2.5 specification, for use case notation.

2 Overall Description

2.1 Product Overview

CGAS is a new, self-contained web application for the campus. It does not replace an existing software system; it replaces paper registers, phone calls and word-of-mouth complaints with one digital channel.

The system has two front ends on one shared database. The student portal lets a student log in, raise a ticket, and track it. The administration portal gives staff one queue of all tickets, sorted by time, where they assign, update and close them. A deduplication module sits between the two: when a ticket is submitted, it compares the ticket with open tickets of the same category and location, and links likely duplicates to one parent ticket.

Figure 1 shows how the parts connect.

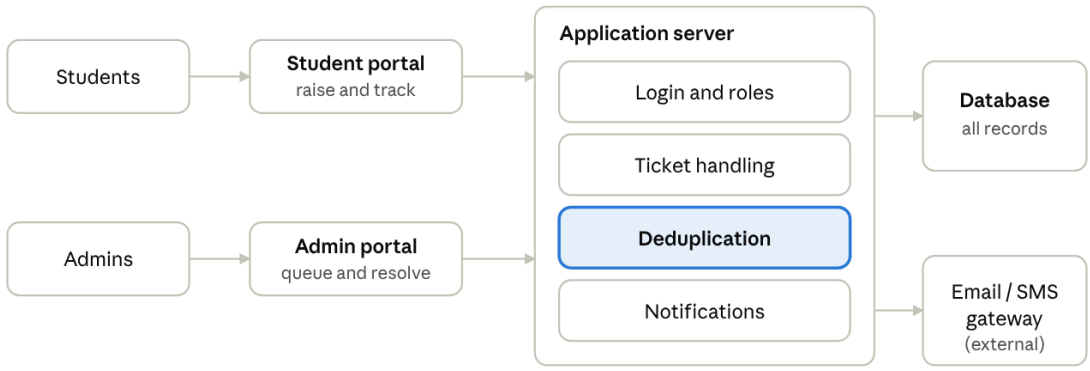


Figure 1: System context diagram of CGAS

Both portals talk only to the application server. The server alone reads the database and calls the email and SMS gateway.

2.2 Product Functionality

- Login with role (Student or Admin) and password change
- Raise a grievance in one of six categories
- Automatic duplicate detection and linking
- Track ticket status and history
- Sort tickets by time (newest or oldest first) and filter by category or status
- Admin view of all tickets with assignment, status update and closure
- Notifications on ticket updates
- Admin management of student records, categories and staff
- Student feedback on resolved tickets

2.3 Design and Implementation Constraints

- The project is an Object Oriented Programming Lab project, so the design shall use object oriented principles (classes for User, Ticket, Category, Notification and so on) in an object oriented language chosen with the project owner.

- The system shall run in current versions of Chrome, Firefox and Edge, and be usable on a phone screen.
- Data shall be stored in a relational database (for example MySQL or SQLite).
- Only registered students and administrators may log in, so the system depends on a student and staff record list supplied by the administration.
- Passwords shall be stored hashed, never as plain text.
- Development is limited to one academic semester by a team of four.

2.4 Assumptions and Dependencies

- Students and administrators have a campus email address or phone number registered in the system.
- Administration provides the initial student and staff records before launch.
- The campus network gives reliable access to the server.
- An email or SMS gateway is available for OTPs and notifications. If not, notifications appear only inside the portal.
- Each grievance belongs to exactly one category, and the six categories listed in the scope are enough at launch.
- Administrators will check the administration portal regularly and pass assigned work on to maintenance and administrative staff.

3 Specific Requirements

3.1 Functional Requirements

The system has 15 functional requirements, F1 to F15, which expand the functions listed in section 2.2.

F1: The system shall provide a login page with a role selection (Student / Admin) and authenticate users by ID and password.

F2: The system shall show an alert when a wrong password is entered and lock the account for 15 minutes after 5 failed attempts.

F3: The system shall allow a user to change their password after verifying an OTP sent by email or SMS.

F4: The system shall allow a student to raise a ticket by choosing a category (Plumbing, Carpentry, Electrical, Masonry, Administrative, Security), a location, a description, an optional photo and an optional confidential marking that makes the ticket visible only to admins.

F5: The system shall record a timestamp and a unique ticket ID for every ticket and set its status to OPEN.

F6: The system shall compare each new ticket with open tickets of the same category and location and warn the student of possible duplicates before submission.

F7: The system shall let the student either support an existing ticket or submit anyway. A supported ticket shall be linked to the existing (parent) ticket and marked DUPLICATE; a ticket submitted anyway shall be created as OPEN and flagged to the admin as a possible duplicate.

F8: The system shall allow a student to view all their tickets with current status and the history of updates.

F9: The system shall allow an admin to view all tickets in one unified queue and to search it by ticket ID.

F10: The system shall sort tickets by time (newest first or oldest first) and shall allow filtering by category, status and location.

F11: The system shall allow an admin to assign a ticket to a staff member, change its status (OPEN, IN PROGRESS, RESOLVED, CLOSED) and add a remark. Closing a parent ticket shall also close all tickets linked to it as duplicates.

F12: The system shall notify a student when the status of their ticket changes.

F13: The system shall allow a student to give a rating and feedback on a RESOLVED ticket and to reopen it within 7 days if the problem persists.

F14: The system shall allow an admin to add student and staff records (one by one or by uploading a CSV file of student records) and to add, edit or remove grievance categories.

F15: The system shall allow an admin to view a log of all ticket actions and to export a summary of tickets by category and status.

3.2 Use Case Model

Figure 2 shows the use case diagram of the system.

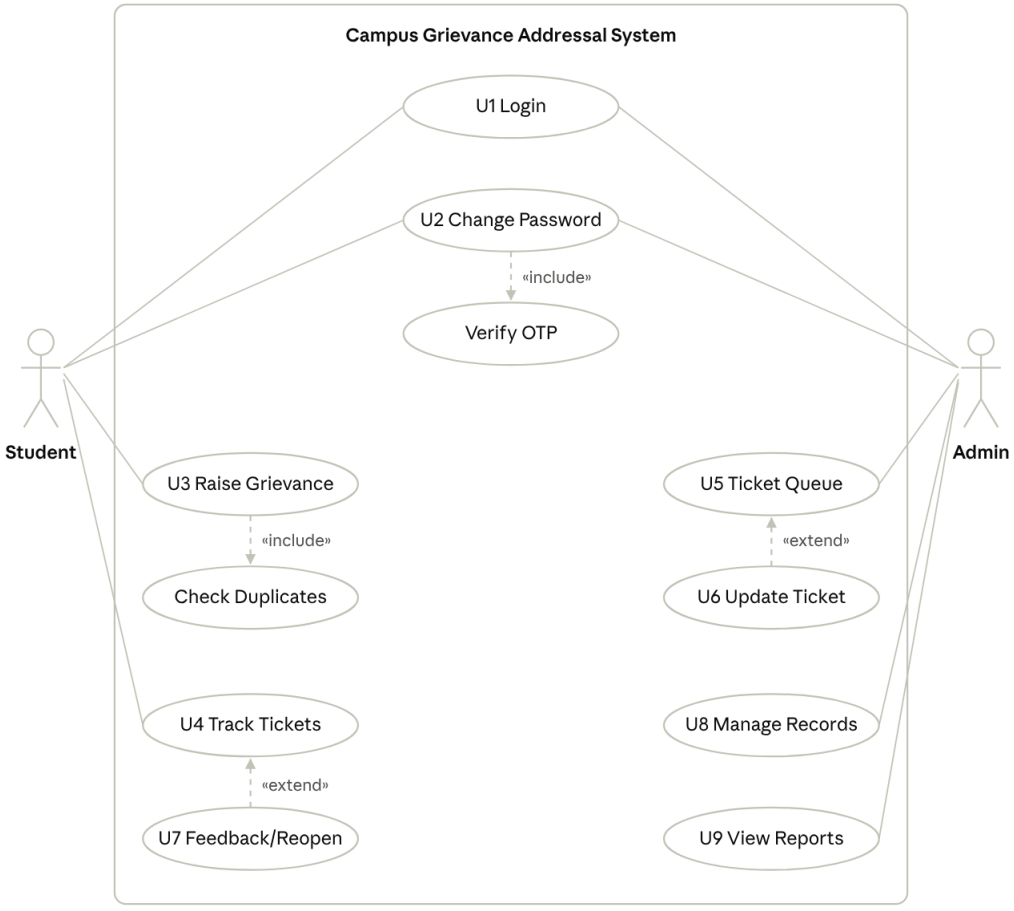


Figure 2: Use case diagram of CGAS

Students own U3, U4 and U7; admins own U5, U6, U8 and U9; both share Login and Change Password. Verify OTP and Check Duplicates are included use cases, described within U2 and U3. Dashed arrows show include and extend relations, drawn in UML 2.5 notation.

3.2.1 Use Case #1 (Login – U1)

Author	Jibendra Kumar Garai
Purpose	Authenticate a user and open the portal for their role
Requirements Traceability	F1, F2
Priority	High
Preconditions	The user has a registered account
Post conditions	The user is logged in and sees the home page for their role
Actors	Student, Admin

Extends	None
Basic Flow	1. User opens the login page. 2. User selects role and enters ID and password. 3. System verifies the credentials. 4. System opens the role's home page.
Alternative Flow	The user chooses Forgot Password and is taken to U2.
Exceptions	Wrong password: system shows an alert and allows a retry. After 5 failures the account is locked for 15 minutes.
Includes	None
Notes/Issues	Passwords are stored hashed.

3.2.2 Use Case #2 (Change Password – U2)

Author	Jibendra Kumar Garai
Purpose	Let a user change their password after verifying an OTP
Requirements Traceability	F3
Priority	Medium
Preconditions	The user has an account with a registered email or phone number
Post conditions	The password is updated and the old one no longer works
Actors	Student, Admin
Extends	None
Basic Flow	1. User selects Change Password. 2. System sends an OTP. 3. User enters the OTP. 4. User enters and confirms a new password. 5. System saves it and confirms.
Alternative Flow	User requests a new OTP if the first has not arrived.
Exceptions	Wrong or expired OTP (valid 5 minutes): system rejects it and keeps the old password.
Includes	Verify OTP
Notes/Issues	None

3.2.3 Use Case #3 (Raise Grievance – U3)

Author	Soham Siddhartha Roy
Purpose	Let a student file a categorised grievance through the single portal

Requirements Traceability	F4, F5, F6, F7
Priority	High
Preconditions	The student is logged in
Post conditions	A ticket with a unique ID and timestamp exists. It is OPEN, or DUPLICATE and linked to a parent ticket if the student supported an existing one
Actors	Student
Extends	None
Basic Flow	1. Student selects New Grievance. 2. Student picks a category and location and writes a description, with an optional photo and an optional confidential marking. 3. System runs Check Duplicates. 4. No match is found, so the system creates the ticket and shows its ID.
Alternative Flow	A likely duplicate is found. The system shows the existing ticket. The student either supports it (the new ticket is marked DUPLICATE and linked) or submits anyway (the ticket is created as OPEN and the admin sees it flagged).
Exceptions	A required field is empty: system highlights it and does not submit. Photo over 5 MB: system rejects it.
Includes	Check Duplicates
Notes/Issues	Duplicate matching uses category, location and text similarity on open tickets only.

3.2.4 Use Case #4 (Track Tickets – U4)

Author	Soham Siddhartha Roy
Purpose	Let a student see the status and history of their grievances
Requirements Traceability	F8, F12
Priority	High
Preconditions	The student is logged in and has at least one ticket
Post conditions	None; no data changes
Actors	Student
Extends	None
Basic Flow	1. Student opens My Tickets. 2. System lists tickets, newest first. 3. Student opens a ticket and sees status, assignee, remarks and timeline.
Alternative Flow	The student sorts the list oldest first or filters by status.

Exceptions	No tickets exist: system shows an empty-state message.
Includes	None
Notes/Issues	A notification is sent on each status change (F12).

3.2.5 Use Case #5 (View and Sort Ticket Queue – U5)

Author	Chaitanya Sathone
Purpose	Give the admin one time-sorted queue of all tickets from all categories
Requirements Traceability	F9, F10
Priority	High
Preconditions	The admin is logged in
Post conditions	None; no data changes
Actors	Admin
Extends	None
Basic Flow	1. Admin opens the Ticket Queue. 2. System lists all tickets, oldest unresolved first. 3. Admin changes the sort to newest first or applies filters (category, status, location).
Alternative Flow	The admin searches by ticket ID.
Exceptions	No ticket matches the filters: system shows a message and a Clear Filters button.
Includes	None
Notes/Issues	Duplicate tickets are shown grouped under their parent ticket.

3.2.6 Use Case #6 (Update Ticket – U6)

Author	Chaitanya Sathone
Purpose	Let the admin assign, progress and close tickets
Requirements Traceability	F11, F12
Priority	High
Preconditions	The admin is logged in and the ticket exists
Post conditions	The ticket's status, assignee and remark are saved and the student is notified
Actors	Admin

Extends	U5
Basic Flow	1. Admin opens a ticket from the queue. 2. Admin assigns a staff member. 3. Admin sets the status and adds a remark. 4. System saves the change, logs it and notifies the student.
Alternative Flow	Closing a parent ticket also closes all linked duplicate tickets.
Exceptions	Two admins edit the same ticket: the system rejects the second save and asks for a refresh.
Includes	None
Notes/Issues	None

3.2.7 Use Case #7 (Give Feedback or Reopen – U7)

Author	Aniket Rajesh Borkar
Purpose	Let a student rate a resolved ticket or reopen it if the problem remains
Requirements Traceability	F13
Priority	Medium
Preconditions	The ticket is RESOLVED
Post conditions	The rating is saved, or the ticket returns to OPEN
Actors	Student
Extends	U4
Basic Flow	1. Student opens a RESOLVED ticket. 2. Student gives a rating (1 to 5) and a comment. 3. System saves it.
Alternative Flow	Student selects Reopen within 7 days and the ticket returns to OPEN with a note.
Exceptions	More than 7 days have passed: the system disables Reopen and asks the student to raise a new ticket.
Includes	None
Notes/Issues	None

3.2.8 Use Case #8 (Manage Records and Categories – U8)

Author	Aniket Rajesh Borkar
Purpose	Let the admin maintain student records, staff records and grievance categories

Requirements Traceability	F14
Priority	Medium
Preconditions	The admin is logged in
Post conditions	The records or categories are added, edited or removed
Actors	Admin
Extends	None
Basic Flow	1. Admin opens Manage Records. 2. Admin adds or edits a student, staff member or category. 3. System validates and saves it.
Alternative Flow	Admin uploads a CSV of student records.
Exceptions	A duplicate student ID is entered: system rejects it. A category with open tickets cannot be deleted.
Includes	None
Notes/Issues	None

3.2.9 Use Case #9 (View Logs and Reports – U9)

Author	Jibendra Kumar Garai
Purpose	Let the admin audit ticket actions and see counts by category and status
Requirements Traceability	F15
Priority	Low
Preconditions	The admin is logged in
Post conditions	None; no data changes
Actors	Admin
Extends	None
Basic Flow	1. Admin opens Reports. 2. Admin picks a date range. 3. System shows the log and a summary by category and status. 4. Admin exports it.
Alternative Flow	None
Exceptions	No data in the range: system shows an empty report.
Includes	None
Notes/Issues	None

4 Other Non-functional Requirements

4.1 Performance Requirements

- Pages shall load in under 3 seconds on the campus network with up to 200 users online at once.
- The duplicate check (F6) shall return its result within 2 seconds of the student finishing the description.
- The ticket queue shall sort or filter up to 10,000 tickets in under 2 seconds.
- The system shall be available 95% of the time, excluding planned maintenance.
- Notifications shall be sent within 1 minute of a status change.

4.2 Safety and Security Requirements

- Users shall authenticate before any ticket data is shown, and each role shall see only its own pages. A student can see only their own tickets; an admin can see all.
- Passwords shall be stored using a salted hash. The session shall expire after 15 minutes of inactivity.
- OTPs shall be valid for 5 minutes and usable once.
- All input shall be validated on the server to prevent SQL injection and cross-site scripting.
- Uploaded photos shall be limited to JPG and PNG up to 5 MB and checked by file type, not trusted by file name.
- Security grievances may contain sensitive details, so the system shall allow a student to mark a ticket confidential (F4), visible only to admins.
- Student personal data shall be used only for grievance handling and shall not be shared outside the institution.
- Every ticket action shall be logged with the user and time, and the log shall not be editable from the portal.

5 Other Requirements

- Usability: a student shall be able to raise a ticket in at most 4 steps from the home page.
- Maintainability: the code shall be modular, with one class per entity, and each class shall be documented.
- Portability: the system shall run on Windows and Linux servers.

Appendix A - Activity Log

Date	Activity	Members	Hours
27/09/2026	Deciding the project idea and features	Aniket, Soham, Jibendra, Chaitanya	3
04/10/2016	Research	Soham, Aniket, Jibendra, Chaitanya	6
09/10/2026	Drafting and reviewing the SRS	Aniket, Soham	4